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Invert4TVG: A Temporal Video Grounding Framework with Inversion Tasks for Enhanced Action Understanding

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Abstract

Temporal Video Grounding (TVG) seeks to localize video segments matching a given textual query. Current methods, while optimizing for high temporal Intersection-over-Union (IoU), often overfit to this metric, compromising semantic action understanding in the video and query, a critical factor for robust TVG. To address this, we introduce *Inver*sion *T*asks for *TVG* (Invert4TVG), a novel framework that enhances both localization accuracy and action understanding without additional data. Our approach leverages three inversion tasks derived from existing TVG annotations: (1) Verb Completion, predicting masked action verbs in queries from video segments; (2) Action Recognition, identifying query-described actions; and (3) Video Description, generating descriptions of video segments that explicitly embed query-relevant actions.These tasks, integrated with TVG via a reinforcement learning framework with well-designed reward functions, ensure balanced optimization of localization and semantics. Experiments show our method outperforms state-of-the-art approaches, achieving a 7.1% improvement in R1@0.7 on Charades-STA for a 3B model compared to Time-R1. By inverting TVG to derive query-related actions from segments, our approach strengthens semantic understanding, significantly raising the ceiling of localization accuracy.

Introduction

Temporal Video Grounding (TVG) is crucial for long-form video understanding (Gaidon, Harchaoui, and Schmid 2013; Laptev and Pérez 2007; Darrell and Pentland 1993). It localizes a video segment matching a textual query (Gao et al. 2017; Zhang et al. 2023), enabling applications like video-text retrieval (Zhang 2024) and UAV positioning (Ju 2024).

Existing TVG approaches fall into three paradigms: (1) Traditional methods using hand-crafted features, sliding windows, and DETR-like networks (Shi et al. 2022; Gordeev 2024); (2) LVLMs (Bai et al. 2025; Li et al. 2023) that regress segments via pretraining; and (3) RL-finetuned LVLMs, such as TimeZero (Xu 2025b) and Time-R1 (Xu 2025a), which use RL with format rewards for structured reasoning and IoU rewards for alignment.

TVG is fundamentally a video understanding task, with a particular emphasis on comprehension of actions, since textual queries typically contain verbs describing human activities (e.g., "opening a door"). These action-related verbs are critical for aligning video content with the query intent.

However, existing TVG methods primarily optimize for localization metrics like Intersection-over-Union (IoU), which measure the temporal overlap between predicted and ground-truth segments. However, models may overfit to superficial localization cues (e.g., relying on visual motion patterns rather than action semantics) and neglect deeper alignment between video actions and query verbs, leading to superficial matching rather than true understanding.

Figure 1 demonstrates this. Time-R1 (Xu 2025a), optimized for IoU in TVG, shows superior performance on TVG metrics (e.g., R1@0.3/R1@0.5/R1@0.7) compared to the baseline Qwen2.5-VL-3B. However, it exhibits degradation in key action-centric understanding tasks (VC/VD/AR), which *ultimately* hampering the TVG accuracy.

A key insight of our work is that improving action-level semantics is essential for pushing the upper bound of TVG accuracy. Conventional solutions involve joint training TVG tasks with other video understanding tasks such as VQA or video captioning. This is exactly how existing large vision-language models (LVLMs), e.g., Qwen2.5-VL series, are trained. However, this paradigm requires data expansion. More critically, VQA and captioning tasks are not originally designed to address action understanding in TVG scenarios. The research question in this paper is: *Can we enhance TVG while preserving video-query action comprehension within existing TVG resources?*

We propose Invert4TVG, a novel framework that derives auxiliary action understanding tasks from TVG datasets without requiring external data or annotations. Specifically, we invert the input and output of the TVG task to obtain Invert-TVG tasks. Originally, given a video and a natural language query, TVG models predict the temporal segment where the action occurs. Inversely, given a video segment, Invert-TVG tasks infer the action-related information defined in the query. We introduce three Invert-TVG tasks: (1) Verb Completion (VC): Mask and infer action verbs from video segments. (2) Action Recognition (AR): Identify query actions from the segment. (3) Video Description (VD): Generate segment descriptions that contain action verbs in the queries.

With the well-defined Invert-TVG tasks, we then pro-

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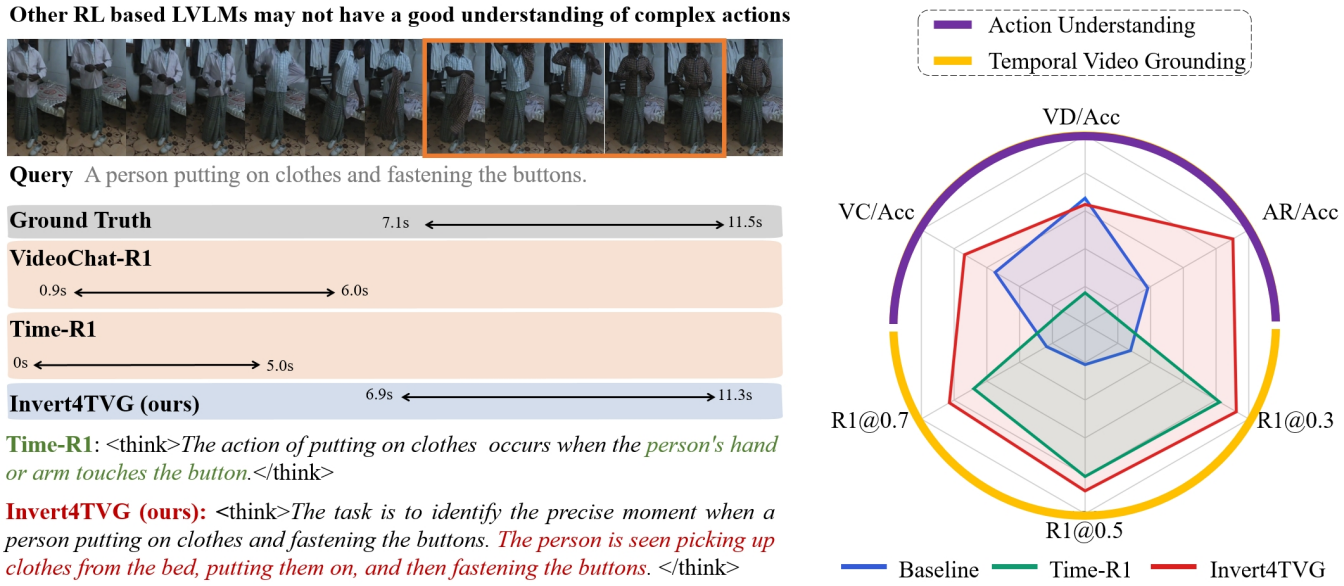

Figure 1: **Left side:** A specific example of temporal video grounding. Through the model's reasoning process, it can be observed that our method achieves a better understanding of actions in the video compared to Time-R1 (Xu 2025a) and other approaches. **Right side:** statistical results demonstrating that Time-R1, which is optimized solely for IoU loss, improves localization accuracy but reduces action understanding accuracy (where VC, AR, and VD are three Invert-TVG tasks proposed in this paper). By introducing Invert-TVG tasks, our method not only enhances action understanding but also boosts the accuracy of TVG (as shown in R1@0.3, R1@0.5, and R1@0.7).

pose a reinforcement learning framework that executes TVG with high probability and samples Invert-TVG tasks equally, balancing localization and action understandings. Figure 1 demonstrates that our method, Invert4TVG, effectively preserves action understanding capabilities, thereby achieving superior performance on the TVG task compared to Time-R1, which solely optimizes the TVG objective without explicit action understanding. Our contributions include:

- We identify action understanding degradation in TVG from metric over-optimization, and address the problem via inversion tasks without extra data.
- We design Invert-TVG tasks which are self-supervised tasks repurposing TVG annotations for action understanding (including verb completion, action recognition, and video description).
- We propose a multi-task reinforcement learning framework dynamically balancing TVG and Invert-TVG, ensuring robust localization and understanding.

Related Works

Temporal Video Grounding. Temporal Video Grounding (TVG) (Gao et al. 2017; Hendricks et al. 2017) localizes specific segments in untrimmed videos based on natural language queries. Recent methods fall into two categories: feature-based and frame-based LVLM approaches. Feature-based methods (Carreira and Zisserman 2017; Lin et al. 2022) extract video and text features using pretrained encoders, then predict timestamps via multimodal fusion. These rely heavily on feature quality, limiting performance. Frame-based LVLM methods (Li et al. 2024; Ren et al. 2024; Zeng et al. 2025) use supervised fine-tuning with autoregressive losses to generate event sequences, but often underperform on benchmarks like Charades-STA compared to feature-based approaches. To address this, Time-R1 (Xu 2025a) employs reinforcement learning (RL) with IoU rewards, achieving state-of-the-art TVG performance. However, its focus on temporal metrics neglects semantic alignment, constraining long-form video understanding

RL in LVLMs. RL has advanced post-training of LVLMs through Reinforcement Learning with Human Feedback (RLHF) (Ouyang et al. 2022; Yu et al. 2024) and Reinforcement Learning with Verifiable Reward (RLVR) (DeepSeek-AI 2025; Chen et al. 2025). RLHF aligns models with human preferences, improving tasks like image captioning, while RLVR enhances deterministic tasks like visual grounding (Liu et al. 2025). However, RL applications in long-form video tasks remain underexplored due to temporal complexity and semantic challenges. TimeZero (Xu 2025b), Time-R1 (Xu 2025a), VideoChat-R1 (Li et al. 2025) apply RL to TVG but overlooks semantic understanding degradation from IoU-focused rewards. Our Invert4TVG framework addresses this by repurposing TVG data into self-supervised tasks, enhancing action semantic understanding and surpassing traditional RL limitations in video grounding.

Method

The TVG task aims to temporally localize video segments within long-form videos based on natural language queries. Given a video V, and a language query q, the goal is to iden-

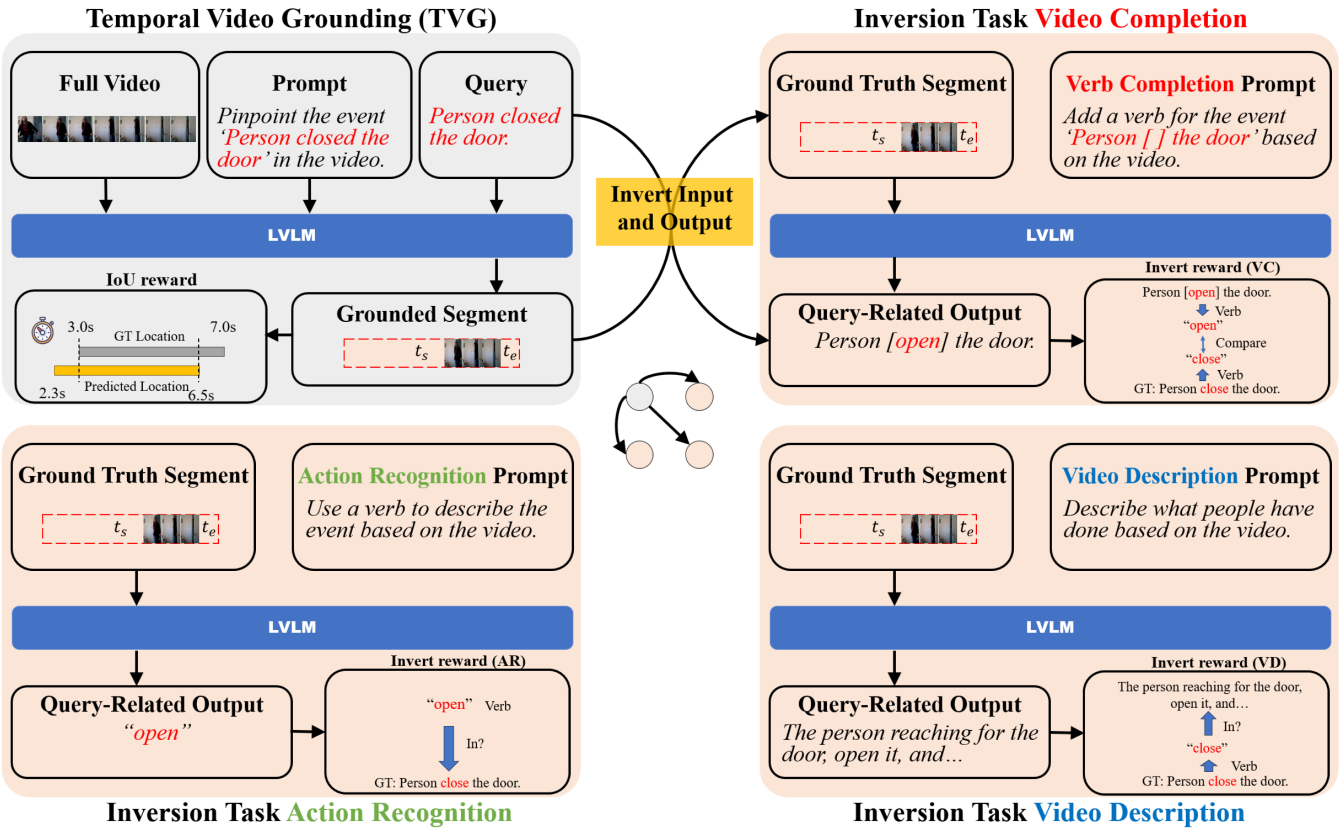

Figure 2: We propose three Invert-TVG tasks. By partially reversing the inputs and outputs of the TVG task we obtain Verb Completion, Action Recognition and Video Description, which reuse the original TVG dataset by taking ground truth video segments as input to reconstruct the target query related events. The prompts for the three invert-TVG tasks are not identical. For VC, the verb in the query is removed, and the model is required to complete and fill in this verb. AR asks the model to directly estimate the verb in the video. VD requires the model to describe the video content containing action verbs in the query.

tify the temporal boundaries $\tau = [t_s, t_e]$ of the segment that best corresponds to q, where $t_s, t_e \in \mathbb{R}^+$. The formal definition of the TVG task is as follows:

$$\text{TVG}(V, q) \rightarrow \tau. \tag{1}$$

In this work, we introduce Invert4TVG, a framework designed to harness the potential of Large Vision-Language Model (LVLM) for the TVG task using Reinforcement Learning (RL) combined with TVG-inversion tasks. The Invert-TVG task is defined as (where q' denotes query-related content):

$$\text{Invert-TVG}(V, \tau) \rightarrow q' \tag{2}$$

Our approach is fundamentally a reinforcement learning algorithm that fine-tunes LVLMs (specifically the Qwen2.5-VL model series) by integrating both TVG and Invert-TVG tasks. In the following, we first introduce the fundamentals of GRPO (i.e., Group Relative Policy Optimization, a reinforcement learning algorithm proposed in (DeepSeek-AI 2025)). Next, we introduce the proposed inversion TVG tasks, together with reward functions used to train the TVG and Invert-TVG tasks. Finally, we introduce our Invert4TVG reinforcement learning framework.

Preliminary of Group Relative Policy Optimization

DeepSeek-R1 (DeepSeek-AI 2025), an early R1-style open-source LLM, uses GRPO to train policy π_θ for reasoning before answers, suitable for precise tasks like math. For query q, it generates responses $o_1, \ldots, o_G$ with score with $r(\cdot)$, and maximizes:

$$R(o) = \sum_{i=1}^{G} \frac{\pi_\theta(o_i)}{\pi_{\theta_{\text{old}}}(o_i)} \cdot \frac{r(o_i) - \text{mean}\left(\{r(o_i)\}_{i=1}^{G}\right)}{\text{std}\left(\{r(o_i)\}_{i=1}^{G}\right)}, \tag{3}$$

where $\pi_\theta(o_i)$ is generation probability, $\pi_{\theta_{\text{old}}}$ is prior state. The full objective with KL is:

$$\max_{\pi_\theta} \mathbb{E}_{o \sim \pi_{\theta_{\text{old}}}(p)} \left[R(o) - \beta \mathrm{D}_{\text{KL}}\left(\pi_\theta \| \pi_{\text{ref}}\right)\right], \tag{4}$$

where β is a scaling coefficient. We omit the clipping operation for simplicity.

Invert-TVG Tasks and Reward Functions

In Temporal Video Grounding (TVG), the accuracy of model inference highly depends on its understanding of both the video V and the query q. Therefore, we do not rely solely on the IoU reward but also introduce additional rewards to

enhance or at least keep the model's action understanding capability. To this end, as illustrated in Figure 2, we design three Invert-TVG tasks and define reward functions measuring how these tasks are fulfilled.

Verb Completion. Verb Completion is a task that masks verbs in the query and asks the model to recover the verbs from the ground truth video segment. For example, if the original query is "Person closed the door", the masked sentence is "Person [ ] the door". The prompt is "Add a verb for the event 'Person [ ] the door' based on the video". As long as the model outputs a sentence successfully recovering the verbs of the ground truth, a reward is given. For instance, if the output sentence is "Person [closes] the door" a full reward is given. Due to the randomness in the model's output, we are primarily concerned with whether the model comprehends the actions within the relevant segments. Therefore, we design the system to treat verbs in different tenses as equivalent. The reward function is as follows:

$$\mathrm{r}_{\mathrm{VC}}(o) = \begin{cases} 0 & v_{\text{pred}} \neq v_{\text{gt}} \\ 1 & v_{\text{pred}} = v_{\text{gt}} \\ 1 & v_{\text{pred}} = v_{\text{variant}} \end{cases} \tag{5}$$

where v_{pred} represents the predicted verb in the output sentence o, and v_{gt} represents the verb in the ground truth sentence. A full reward is obtained if they are equal, or if v_{pred} is a variant $v_{variant}$ of v_{gt}, the same full reward is given too.

Action Recognition. Action Recognition task trains and preserves the model's action perception capability, and the output of the model is fixed to a single verb. We feed the model the ground truth video segment with prompt as "Use a verb to describe the event based on the video", then compare the predicted verb against any verbs appearing in the ground truth query description. If the model's predicted verb is present in the reference sentence, it receives a full reward. For example, if the model outputs "walk" and the ground-truth sentence is "A person walks away and laughs", a full reward is granted because "walk" occurs in the reference. As before, verb in different tenses are treated as equivalent. The reward function is as follows:

$$\mathrm{r}_{\mathrm{AR}}(o) = \begin{cases} 0 & v_{\text{pred}} \notin S_{\text{gt}} \\ 1 & v_{\text{pred}} \in S_{\text{gt}} \\ 1 & v_{\text{variant}} \in S_{\text{gt}} \end{cases} \tag{6}$$

where v_{pred} represents the predicted verb o, and S_{gt} is the set of verbs in the ground truth sentence. A full reward is given if v_{pred} is present in the set of verbs in S_{gt}, or if a variant $v_{variant}$ of v_{pred} exists within the set of verbs in S_{gt}.

Video Description. Video Description task is employed to train and maintain the model's holistic perception of events, yielding a complete segment level description. Specifically, we feed the model the ground-truth video segment and prompt as "Describe what people have done based on the video". A full reward is granted as long as ground-truth verbs appear in the output sentence of the model. For instance, if the ground-truth verb is "jump" and the model produces "A person jumps and laughs" the reward is awarded because "jump" is present. The reward function is as follows:

$$\mathrm{r}_{\mathrm{VD}}(o) = \begin{cases} 0 & v_{\text{gt}} \notin S_{\text{pred}} \\ 1 & v_{\text{gt}} \in S_{\text{pred}} \\ 1 & v_{\text{variant}} \in S_{\text{pred}} \end{cases} \tag{7}$$

where v_{gt} represents the ground truth verb, and S_{pred} denotes the set of verbs in the sentence o predicted by the model.

IoU and Format Reward Functions

For the TVG task, we mainly employ the IoU reward function. Besides, we introduce a Format reward to enforce the model to output the thinking process.

IoU Reward. As stated above, TVG aims at estimating the time interval in the video that is associated with the content of a given textual query. We use the Intersection over Union (IoU) (Yuan et al. 2021) between the time interval predicted by the model and the ground-truth interval as the reward function. This reward function effectively describes the accuracy of the time interval predicted by the model. Given predicted $[t_s, t_e]$ and ground-truth $[t'_s, t'_e]$ segments, the IoU reward can be calculated as follows:

$$r_{\mathrm{IoU}}(o) = \frac{|[t_s, t_e] \cap [t'_s, t'_e]|}{|[t_s, t_e] \cup [t'_s, t'_e]|} \tag{8}$$

where $\cap$ and $\cup$ denote set intersection and union operations.

Format Reward. Recently, Time-R1 (Xu 2025a) and VideoChat-R1 (Li et al. 2025) employ a Format reward making the model explicitly output its thinking process before making predictions. Following them, we introduce a template-based reasoning reward that incentivizes the model to generate intermediate reasoning steps prior to providing answers. The format is as following: `<think>***</think> <answer>` t_s to t_e `</answer>`. The reward is formulated as:

$$r_{\mathrm{form}}(o) = \begin{cases} 0, & \text{if } o \text{ has wrong format} \\ 1, & \text{if } o \text{ has correct format} \end{cases} \tag{9}$$

Invert4TVG Reinforcement Learning Framework

While a joint training approach that processes all TVG and Invert-TVG tasks simultaneously might seem straightforward, this method suffers from several critical limitations: (1) Memory inefficiency: maintaining separate computation graphs for multiple tasks drastically increases GPU memory consumption; (2) Optimization conflict: gradient updates from different tasks may interfere with each other, especially when their loss landscapes are not aligned; (3) Training instability: the varying convergence rates of different tasks make it challenging to balance their contributions; (4) Task bias: the model may prioritize easier tasks while neglecting others. These drawbacks motivate our proposed alternating training paradigm.

We (see Figure 3) implement a probabilistic sampling strategy where each training iteration has an high probability (80% in this paper) of executing the primary TVG task (using IoU and format rewards) and a low probability of performing an Invert-TVG task. When selecting Invert-TVG, we uniformly sample among VC, AR and VD. This design

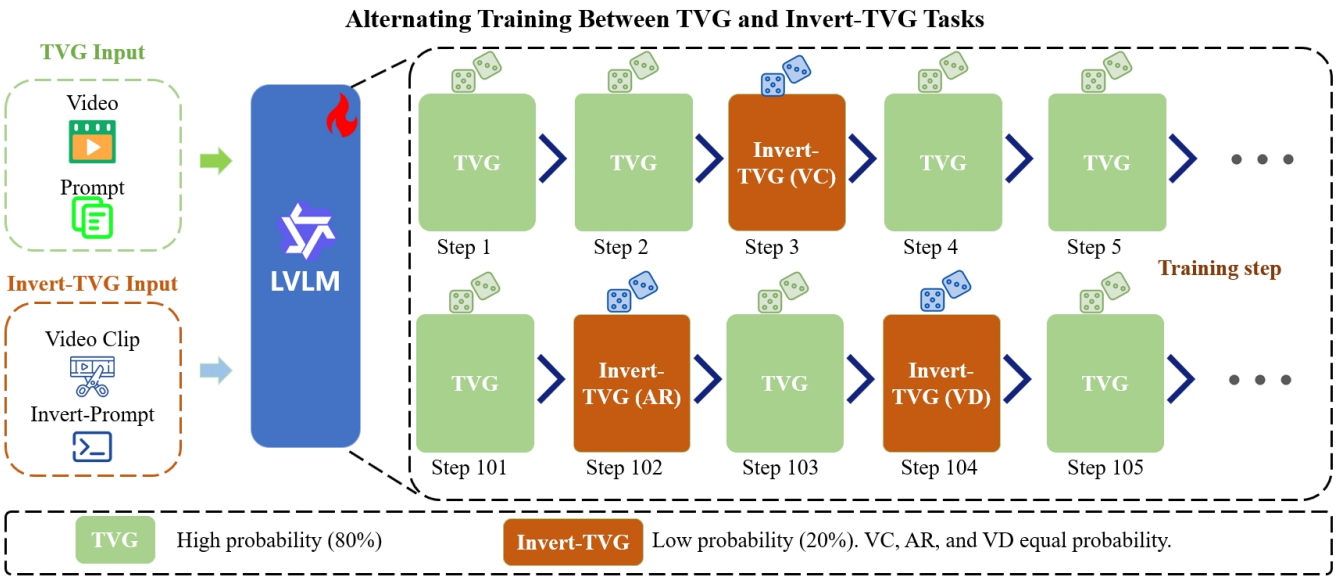

Figure 3: Overview of the proposed Invert4TVG framework. During reinforcement learning training the Invert4TVG prevents conflicts between TVG and Invert-TVG tasks and ensures each can be optimized with independent gradients. The LVLM dynamically chooses between TVG tasks and Invert-TVG tasks according to different probabilities. Whenever an Invert-TVG task is selected, one of the three variants VC, AR or VD is chosen with equal probability.

ensures the model maintains its core action understanding capabilities while primarily focusing on temporal localization. The asymmetric probability distribution prevents the auxiliary tasks from overwhelming the main objective while still providing regular semantic reinforcement. Formally, the reward for training the TVG task is:

$$r_{\text{TVG}}(o) = r_{\text{format}}(o) + r_{\text{IoU}}(o). \tag{10}$$

The reward used to train an Invert-TVG task is:

$$r_{\text{Invert-TVG}}(o) = r_{\text{format}}(o) + r_{\text{inv}}(o), \tag{11}$$

where r_{inv} is any of r_{VC}, r_{AR}, and r_{VD}. The overall reward function is defined as:

$$r(o) = \alpha r_{\text{TVG}}(o) + \beta r_{\text{Invert-TVG}}(o), \tag{12}$$

where the two coefficients α and β take values in $\{0, 1\}$, with the constraint $\alpha + \beta = 1$. Their joint probability distribution is defined as (where $0 \leq p \leq 1$):

$$P(\alpha, \beta) = \begin{cases} p & \text{if } (\alpha, \beta) = (1, 0), \\ 1 - p & \text{if } (\alpha, \beta) = (0, 1), \\ 0 & \text{otherwise.} \end{cases} \tag{13}$$

As stated above, p is 0.8 in this paper.

Experiments

We now evaluate our Invert4TVG model on the task of temporal video grounding. Code is attached.

Experimental Setup

Benchmarks. We test our model on three temporal video grounding datasets: (1) Charades-STA (Sigurdsson et al. 2016) contains 6,672 long videos capturing indoor human activities. The official split for the TVG task includes 12,408 clip-query pairs for training and 3,720 for testing. (2) ActivityNet (Heilbron et al. 2015) comprises 20K long videos with an average of 3.65 clip-query pairs per video. We use the standard dataset splits with 37,421 training, 17,505 validation, and 17,031 test samples. (3) We further evaluate on QvHighlight (Lei, Berg, and Bansal 2021), a high-resolution set of 10,460 long YouTube videos paired with 48k manually annotated clip-queries. To match Charades-STA and ActivityNet formats, multi-segment localizations are split into single-segment tasks, forming a balanced benchmark for fine-grained temporal grounding.

Implementation Details We implement our LVLM using the Qwen2.5-VL model (Bai et al. 2025) as the backbone. To balance efficiency and memory consumption, we sample video frames at 2 FPS and resize them, resulting in approximately 2.8 million pixels per video (e.g., a 50-second video yields 100 frames of size $96 \times 96 \times 3$). Our implementation utilizes SpaCy's `en_core_web_sm-3.8.0` model (12MB) to extract verbs from sentences. For optimization, we employ the AdamW optimizer (Loshchilov and Hutter 2019) with the following parameters: $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1 \times 10^{-8}$, a weight decay of 0.0, and a learning rate of 5×10^{-5}. The training time per epoch is approximately 80 hours. To ensure reproducibility, our code, configuration files, and execution scripts are available in the supplementary materials.

Evaluation metrics. For TVG, we adopt the "R1@m" evaluation protocol to compare with state-of-the-art models, which computes the percentage of samples where the top-1 predicted segment has an IoU greater than a threshold m,

Type	Method	Size	FT	Charades-STA		
				R1@0.3	R1@0.5	R1@0.7
VLP	2D-TAN*	-	✓	57.3	45.8	27.9
	Moment-DETR*	-	✓	65.8	52.1	30.6
	EaTR*	-	✓	-	68.4	44.9
	SnAG*	-	✓	-	64.6	46.2
SFT	VideoChat-Flash	7B		74.5	53.1	27.6
	TRACE	7B		-	40.3	19.4
	HawkEye*	7B	✓	72.5	58.3	28.8
	TimeSuite*	7B	✓	79.4	67.1	43.0
RL(3B)	Time-R1(3B)	3B		74.6	53.1	26.0
	Time-R1*(3B)	3B	✓	78.7	64.1	36.9
	Invert4TVG (ours 3B)	3B	✓	**80.8**	**69.0**	**44.0**
RL(7B)	Time-R1 (7B)	7B		78.1	60.8	35.3
	Time-R1*(7B)	7B	✓	82.8	72.2	50.1
	Invert4TVG (ours 7B)	7B	✓	**83.0**	**72.5**	**51.4**

Table 1: Performance of temporal video grounding on Charades-STA. The methods marked with * indicate the use of additional VTG datasets, while FT with ✓indicates training on the Charades-STA training set. We compare our method against existing 3B, 7B open-source LVLM, as well as state-of-the-art VLP models. We highlight our results and the best-performing baselines using bold and underlining for clear comparison.

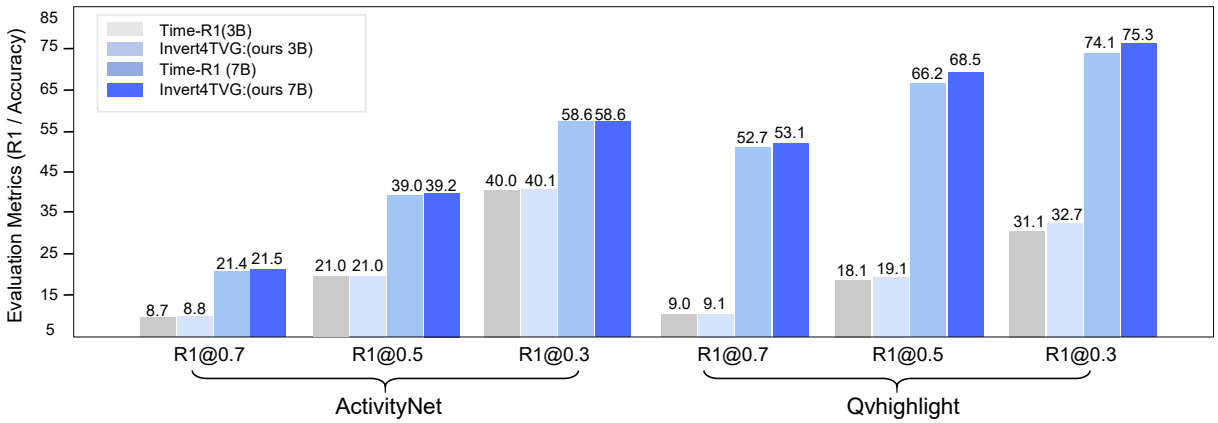

Figure 4: Performance of temporal video grounding on ActivityNet and Qvhighlight (RL methods). All the models are Zero-Shot on the test dataset.

with $m \in 0.3, 0.5, 0.7$. For brevity, we also adopt mIoU, which calculates the average IoU on all testing data as an alternative metric.

Comparison with State-of-the-Art

We compare Invert4TVG with state-of-the-art TVG methods, including both traditional video-language pre-training models (VLP), recent large video-language models fine-tuned via SFT and other RL-based approaches.

Invert4TVG surpasses other models on the Charades-STA dataset. As shown in Table 1, Invert4TVG surpasses not only VLP-based and SFT-based models but also outperforms RL-based approaches under identical conditions. For example, on Charades-STA, the 7B variant of Invert4TVG achieves an R1@0.7 of 51.4, exceeding TimeSuite (43.0), SnAG (46.2), and Time-R1 (50.1). The improvements are more pronounced for the 3B variant. Across R1@0.3, R1@0.5, and R1@0.7, Invert4TVG's 3B model consistently outperforms the 3B version of Time-R1.

Invert4TVG surpasses other RL-based models in zero-shot settings. As shown in Figure 4, in zero-shot settings, Invert4TVG's 3B and 7B variants also outperform Time-R1 on R1@0.3, R1@0.5, and R1@0.7 over ActivityNet. On QvHighlight, where we compare single-segment predictions, Invert4TVG consistently outperforms Time-R1 across R1@0.3, R1@0.5, and R1@0.7, demonstrating its superior accuracy in complex scenarios.

Ablation Study

We conduct a detailed ablation on the Invert4TVG-3B model to investigate the contribution of various design strategies.

Using a single Invert-TVG task versus employing them in combination. As shown in Table 2, using VC, AR, or VD alone instead of jointly yields lower performance. Only-VD, which emphasizes contextual understanding, peaks at R1@0.3 but falls short on precise localization. Only-AR,

Type	Method	Charades-STA		
		R1@0.3	R1@0.5	R1@0.7
RL	Only-VD	79.1	64.3	39.4
	Only-AR	78.2	65.2	43.8
	Only-VC	78.8	68.0	42.0
	Invert4TVG	80.8	69.0	44.0

Table 2: Ablation study using only VC (Verb Completion), AR (action recognition), VD (video description), and their mixed usage.

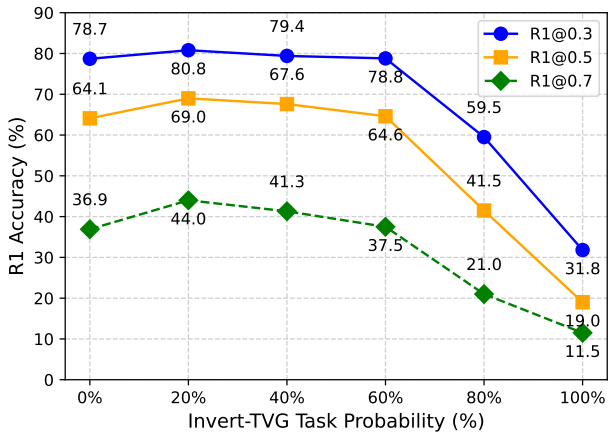

Figure 5: The R1 accuracy curves. Blue, orange, and green show how the three R1 metrics evolve as the Invert-TVG task probability gradually increases.

focused on immediate actions, reaches the highest R1@0.7 of 43.8. Only-VC outputs are less random than Only-VD yet less specific than Only-AR, achieving the best R1@0.5 (68.0). The mixed-task Invert4TVG surpasses all three individual tasks across all three metrics, demonstrating that joint training outperforms separate use.

Exploiting different probabilities for the TVG and Invert-TVG tasks. As shown in Figure 5, Varying the task probability markedly alters training outcomes. At 20% Invert-TVG task probability, the model performs best, raising R1@0.7 from 36.9 (no Invert task) to 44.0. As the Invert-TVG task probability grows, the model increasingly emphasizes action recognition while neglecting temporal grounding. Between 60% and 80% Invert-TVG, temporal video grounding performance steadily declines, falling below the pure TVG baseline. When Invert-TVG probability reaches 100 %, the model performs only the Invert-TVG task and yields the worst results.

Binary Invert-TVG rewards vs. cosine similarity-based rewards. As shown in Table 3, we observe that employing a simple binary Invert-TVG reward (0 or 1) during training yields superior outcomes compared to more intricate reward mechanisms. At identical training steps, the employed Invert-TVG Reward outperforms the cosine similarity reward across all three evaluation metrics (R1@0.3, R1@0.5, R1@0.7). This advantage stems from the controllability and stability of the binary reward design, whereas cosine similarity introduces higher variance and optimization instability. For example, "run" and "eat" yield a cosine similarity of 0.2 despite their weak semantic link. Therefore, binary Invert-TVG reward is a better choice.

Reward	R1@0.3	R1@0.5	R1@0.7
Cosine Similarity	76.2	62.2	39.8
Binary 0 or 1	80.8	69.0	44.0

Table 3: The results using binary Invert-TVG reward or cosine similarity reward for training.

Discussion: Why Invert-TVG Tasks? Is there any alternative?

While our framework proposes three Invert-TVG tasks (VC, AR, VD), other candidates could theoretically enrich video-text understanding. For instance, contrastive learning between video segments and queries could enforce global alignment, or masked frame modeling might enhance temporal reasoning. However, these alternatives lack the task-specific synergy and semantic granularity that make VC, AR, and VD indispensable for TVG. In our method, VC, AR, and VD are directly derived from the TVG task itself, each engineered to isolate and strengthen a critical dimension of action understanding that directly impacts temporal localization. In essence, VC, AR, and VD are not generic self-supervised tasks but purpose-built derivatives of the TVG inverse problem. They distill the essence of action understanding from the TVG task itself, ensuring immediate and targeted improvements in localization accuracy. While contrastive learning or masked modeling may enhance broad semantic capabilities, their indirect impact on TVG-specific action-query alignment lacks the precision and immediacy of our task-specific design.

Conclusion

In this work, we present Invert4TVG, an approach that introduces Invert-TVG tasks, requiring the model to generate query-related content from a video and its ground-truth temporal segment. We design three variants of Invert-TVG, including verb completion, action recognition, and video description. These tasks encourage the model to retain and enhance its action understanding capabilities. We develop a Invert4TVG RL framework that jointly optimizes TVG and Invert-TVG tasks. In addition to standard IoU and format rewards, we introduce Invert-TVG rewards to promote performance on Invert-TVG tasks. During training, the model primarily performs TVG at a high probability, while intermittently switching to Invert-TVG tasks at a lower probability. This balanced strategy ensures robust temporal localization while preserving semantic action-verb alignment. Our work bridges TVG-LVLM gap, unlocking higher extensions in traditional tasks. Experiments demonstrate the effectiveness of our method over existing approaches, achieving significant improvements of grounding accuracy. The reasoning process also shows that the proposed method indeed understands actions better than compared approaches.

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